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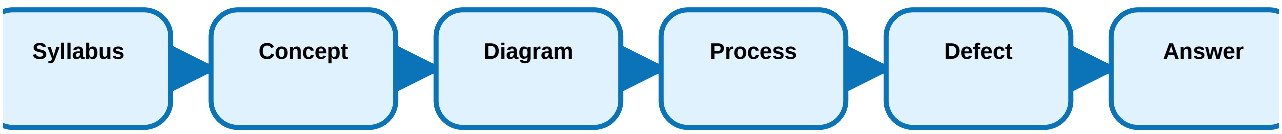
REVISION 2021 • SEMESTER 4 • PROGRAM CORE

Mechanism of Printing Machines II

A complete diploma engineering handbook for Course Code **4101**. This offline lesson converts the official syllabus into a professional study, revision, practical and exam-preparation textbook.

COURSE CODE **4101**	PROGRAMME **Diploma in Printing Technology**
SEMESTER **4**	CREDITS **4**
CATEGORY **Program Core**	PERIODS/WEEK **4 (L:3, T:1 P:0)**
PERIODS/SEMESTER **60**	COURSE TITLE **Mechanism of Printing Machines II**

Production-ready learning map



Study each process through definition, principle, machine parts, workflow, substrates/materials, applications and troubleshooting.

COURSE OVERVIEW

What the subject is and why it matters

What it is

Mechanism of Printing Machines II trains students to understand industrial printing technology as a controlled production system, not as isolated definitions.

Industrial relevance

It connects machine selection, workflow planning, substrate/material control, quality inspection, troubleshooting, waste reduction and job delivery.

How to study

For every topic, learn the definition, working principle, labelled diagram, sequence, advantages, limitations, defects and applications.

Course objectives

- To provide clear and sound knowledge in major print production systems.
- To make judgement about the aspect of printing and differentiate each process.
- To impart knowledge for selecting a suitable process for a specific print production job.
- To identify printing-process problems and propose remedies.

Malayalam support: □ subject theory □□□□ □□□□□□ □□□□□□□□□□. □□□ process-□□□□□ real press workflow, machine setting, material selection, defect and remedy □□□ connect □□□□□□□□.

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COURSE OUTCOMES

Official outcome structure

CO	Description	Hours	Level	Engineering meaning
CO1	Explain the flexographic printing process.	15	Understanding	Use the process in production planning, machine operation, quality inspection and troubleshooting.
CO2	Explain and comprehend the gravure printing process.	14	Understanding	Use the process in production planning, machine operation, quality inspection and troubleshooting.
CO3	Explain and comprehend the screen-printing process.	15	Understanding	Use the process in production planning, machine operation, quality inspection and troubleshooting.
CO4	Assess problems, analyse them and give remedies.	14	Understanding	Use the process in production planning, machine operation, quality inspection and troubleshooting.

SOURCE / SYLLABUS INFORMATION

Confirmed official information and limitations

Program	Diploma in Printing Technology
Course code and title	4101 - Mechanism of Printing Machines II
Semester / credits / category	Semester 4 / 4 credits / Program Core
Periods	4 (L:3, T:1 P:0); 60 periods per semester
Exam pattern	Not specified in supplied syllabus/source. The model paper here is a fresh practice paper arranged for full CO coverage.

Prerequisites

Topic	Course code	Course title	Semester
Parts of machine	Not specified in supplied syllabus/source.	Mechanism of printing machine I	1
Cylinder structure	Not specified in supplied syllabus/source.	Mechanism of printing machine I	1

CO-PO mapping

CO	PO	Mapping
CO1	PO1	2
CO2	PO1	2
CO3	PO1	2
CO4	PO1	2

REDESIGNED SYLLABUS ROADMAP

What to draw, explain, calculate and apply

CO1 - Flexographic Printing Process

Hours: 15

Core topics: Principle, advantages, limitations and characteristics of flexography, Infeed, printing, inking, plate carrier, impression cylinder, drying and outfeed sections, Flexo configurations: inline, stack and central impression, Rubber, photopolymer, metal backed and magnetic image carriers.

Draw: process flow / machine unit / troubleshooting chart.

Explain: principle, parts, operation, advantages, limitations and applications.

Exam importance: High. **Common mistake:** Writing only definitions without diagram and sequence.

CO2 - Gravure Printing Process

Hours: 14

Core topics: Principle, advantages, limitations and characteristics of gravure, Infeed, printing, drying and outfeed sections, Gravure cylinder parts: axis, shaft, diameter, circumference and face length, Static and dynamic balancing.

Draw: process flow / machine unit / troubleshooting chart.

Explain: principle, parts, operation, advantages, limitations and applications.

Exam importance: High. **Common mistake:** Writing only definitions without diagram and sequence.

CO3 - Screen Printing Process

Hours: 15

Core topics: Screen printing definition and image transfer principle, Frames: wooden, metal and self-stretching frames, Mesh materials: silk, synthetic and stainless steel, Stencil types: hand-cut, tusche and glue, photographic.

Draw: process flow / machine unit / troubleshooting chart.

Explain: principle, parts, operation, advantages, limitations and applications.

Exam importance: High. **Common mistake:** Writing only definitions without diagram and sequence.

CO4 - Process Selection and Troubleshooting

Hours: 14

Core topics: Procedure for selecting printing process according to job, Workflow in a printing press and workflow diagram, Flexo troubles: tracking, trapping, mottle, striations, weak colour, adhesion, foaming and smearing, Gravure troubles: abrasion, scumming, drying-in, bleeding, haze, pinholes, blocking, static and brittleness.

Draw: process flow / machine unit / troubleshooting chart.

Explain: principle, parts, operation, advantages, limitations and applications.

Exam importance: High. **Common mistake:** Writing only definitions without diagram and sequence.

MODULE 1 • CO1 • 15 HOURS

Flexographic Printing Process

Explain the flexographic printing process.

Module outcome map

Module outcome	Description	Hours	Level
M1.01	Introduction to Flexography presses	3	Understanding
M1.02	Image carrier preparations	4	Understanding
M1.03	Flexo Printing unit	4	Understanding
M1.04	Flexo substrates	4	Understanding

Textbook chapter explanation

This chapter expands every syllabus topic into practical prepress / press / post-press learning points.

Principle, advantages, limitations and characteristics of flexography

Principle, advantages, limitations and characteristics of flexography

Infeed, printing, inking, plate carrier, impression cylinder, drying and outfeed sections

Infeed, printing, inking, plate carrier, impression cylinder, drying and outfeed sections

Flexo configurations

Flexo configurations: inline, stack and central impression

Rubber, photopolymer, metal backed and magnetic image carriers

Rubber, photopolymer, metal backed and magnetic image carriers

Rubber plate, sheet photopolymer and liquid photopolymer preparation

Rubber plate, sheet photopolymer and liquid photopolymer preparation

Ink metering, anilox roller and doctor blade systems

Ink metering, anilox roller and doctor blade systems

Anilox specifications

Anilox specifications: cell count, depth and volume

Substrates

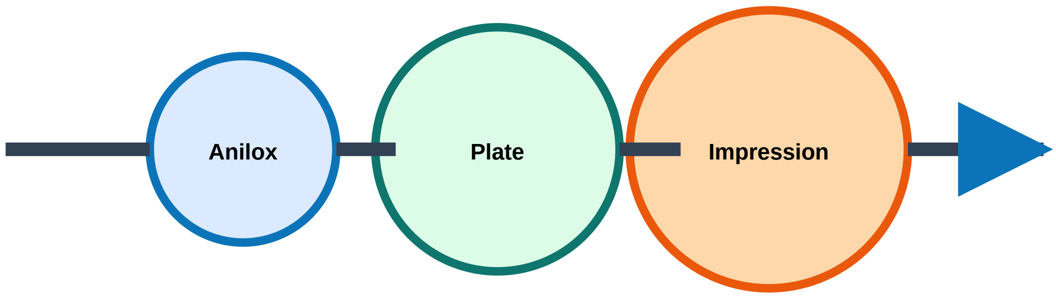
Substrates: paper, paperboard, corrugated board, plastic films, foils and laminates

Corona treatment, wide web, corrugated and narrow web presses, future of flexography

Corona treatment, wide web, corrugated and narrow web presses, future of flexography

Malayalam support: □ module □□□□□□□□□□□□□ definition □□□□□□□ □□□□□□□□□□□□□ □□□□□. Process flow, machine part, job application, defect and remedy □□□□□□□ □□□□ answer-□ □□□□□□□.

Illustration / diagram lab



Ink metering -> plate image -> substrate impression -> drying

Flexographic Printing Process - labelled process diagram for exam answer practice.

Industrial application

Used for labels, flexible packaging, corrugated cartons, foil laminates and narrow web products. In industry, wrong anilox selection gives weak colour, over-inking, dot gain or drying problems.

Exam answer format: definition → principle → labelled diagram → operation sequence → advantages → limitations → applications → defects/remedies.

Troubleshooting table

Problem	Likely cause	Action / remedy
Weak colour	Low ink transfer, wrong anilox, low viscosity	Check ink viscosity, anilox cell volume and doctor blade pressure
Foaming	Air entrainment in ink circulation	Use antifoam, correct pump speed, clean return line
Poor adhesion	Substrate surface energy low	Corona treatment and correct ink system
Smearing	Insufficient drying or excessive ink	Improve dryer, reduce film thickness, check web speed

Module-wise expected questions

- List main sections of a flexographic press.
- Draw a flexo printing unit and label anilox, plate and impression cylinder.
- Explain chambered doctor blade inking system.
- Compare inline, stack and CI flexo presses.
- State the importance of corona treatment.

MODULE 2 • CO2 • 14 HOURS

Gravure Printing Process

Explain and comprehend the gravure printing process.

Module outcome map

Module outcome	Description	Hours	Level
M2.01	Introduction to gravure printing process	3	Understanding
M2.02	Structure of gravure cylinder	2	Understanding
M2.03	Image carrier preparations and printing unit	5	Understanding
M2.04	Drying chamber and substrates	4	Understanding

Textbook chapter explanation

This chapter expands every syllabus topic into practical prepress / press / post-press learning points.

Principle, advantages, limitations and characteristics of gravure

Principle, advantages, limitations and characteristics of gravure

Infeed, printing, drying and outfeed sections

Infeed, printing, drying and outfeed sections

Gravure cylinder parts

Gravure cylinder parts: axis, shaft, diameter, circumference and face length

Static and dynamic balancing

Static and dynamic balancing

Gravure well configuration and effect on tone transfer

Gravure well configuration and effect on tone transfer

Cylinder manufacture

Cylinder manufacture: thin layer, Ballard skin and heavy copper plating

Cylinder preparation

Cylinder preparation: carbon tissue, electromechanical engraving and laser engraving

Doctor blade structure, types, wear, materials and problems

Doctor blade structure, types, wear, materials and problems

Pre-wipe blade, impression roller, electrostatic assist and nib width

Pre-wipe blade, impression roller, electrostatic assist and nib width

Drying chamber, solvent recovery and environment-friendly solvent removal

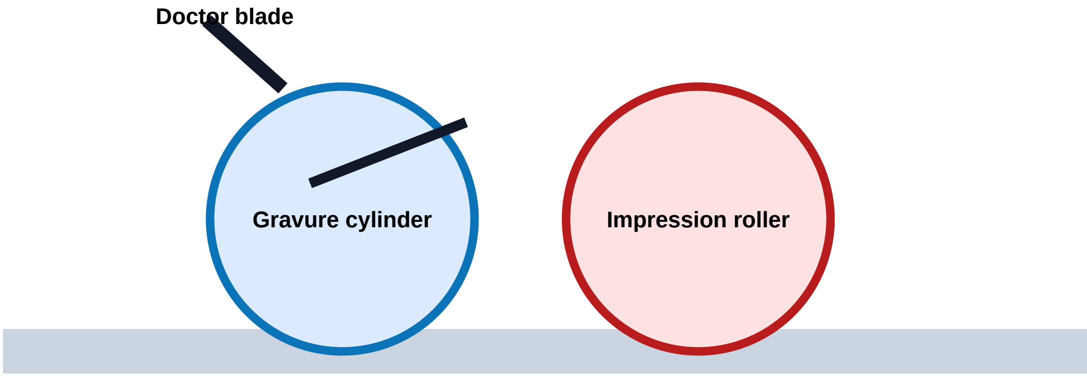
Drying chamber, solvent recovery and environment-friendly solvent removal

Paper and non-paper substrates

Paper and non-paper substrates

Malayalam support: □ module □□□□□□□□□□□□□ definition □□□□□□□ □□□□□□□□□□□□ □□□□□. Process flow, machine part, job application, defect and remedy □□□□□□□ □□□□ answer-□ □□□□□□□.

Illustration / diagram lab



Gravure Printing Process - labelled process diagram for exam answer practice.

Industrial application

Gravure is chosen for long-run high-quality magazines, laminates, packaging and decorative products. Cylinder and doctor blade condition decide consistency, density and streak-free printing.

Exam answer format: definition → principle → labelled diagram → operation sequence → advantages → limitations → applications → defects/remedies.

Troubleshooting table

Problem	Likely cause	Action / remedy
Scumming	Doctor blade not wiping cleanly	Check blade edge, angle, pressure and cylinder polish
Haze	Dirty cylinder or solvent imbalance	Clean cylinder and correct solvent ratio
Pinholes	Ink/substrate wetting issue	Adjust viscosity and substrate treatment
Static	Dry web and high speed	Use antistatic bars and humidity control

Module-wise expected questions

- Draw a gravure printing unit.
- Differentiate static and dynamic balancing.
- Explain gravure cylinder manufacture methods.
- List doctor blade types and problems.
- Explain solvent recovery in gravure drying.

MODULE 3 • CO3 • 15 HOURS

Screen Printing Process

Explain and comprehend the screen-printing process.

Module outcome map

Module outcome	Description	Hours	Level
M3.01	Introduction to screen printing	3	Understanding
M3.02	Types of stencils	4	Understanding
M3.03	Stencil preparations and substrates	4	Understanding
M3.04	Screen printing machines and applications	4	Understanding

Textbook chapter explanation

This chapter expands every syllabus topic into practical prepress / press / post-press learning points.

Screen printing definition and image transfer principle

Screen printing definition and image transfer principle

Frames

Frames: wooden, metal and self-stretching frames

Mesh materials

Mesh materials: silk, synthetic and stainless steel

Stencil types

Stencil types: hand-cut, tusche and glue, photographic

Screen fabrics

Screen fabrics: multifilament and monofilament

Fabric stretching

Fabric stretching: mechanical and hand stretching

Photographic stencil methods

Photographic stencil methods: direct, indirect and direct/indirect

Substrates

Substrates: paper, paperboard, wood, textiles, plastics, metals, ceramics and glass

Squeegee types and selection, on-contact and off-contact printing

Squeegee types and selection, on-contact and off-contact printing

Manual, semi-automatic and automatic machines

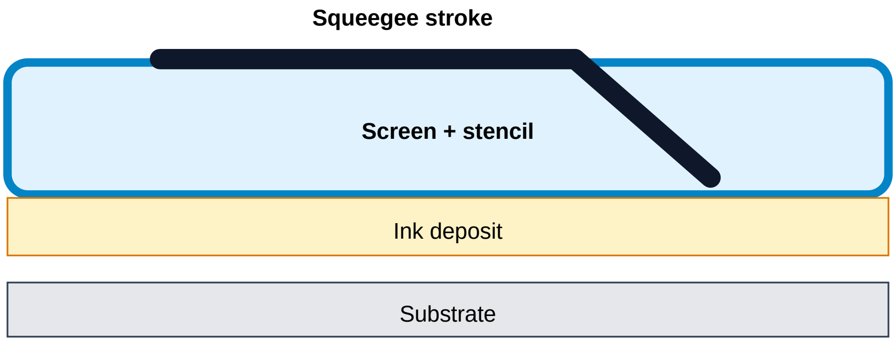
Manual, semi-automatic and automatic machines

Screen printing inks, screen reclamation and applications

Screen printing inks, screen reclamation and applications

Malayalam support: □ module □□□□□□□□□□□□□ definition □□□□□□ □□□□□□□□□□□□ □□□□. Process flow, machine part, job application, defect and remedy □□□□□□□ □□□□ answer-□ □□□□□□□.

Illustration / diagram lab



Screen Printing Process - labelled process diagram for exam answer practice.

Industrial application

Screen printing is used for T-shirts, decals, PCBs, glass, ceramics, panels, name plates and thick ink deposit work. Squeegee hardness, mesh count and off-contact gap decide print quality.

Exam answer format: definition → principle → labelled diagram → operation sequence → advantages → limitations → applications → defects/remedies.

Troubleshooting table

Problem	Likely cause	Action / remedy
Blurred image	Wrong off-contact or low mesh tension	Correct screen tension and off-contact gap
Uneven ink deposit	Wrong squeegee pressure/angle	Set pressure, angle and ink viscosity
Stencil breakdown	Poor exposure or harsh solvent	Correct exposure and use compatible cleaners
Poor registration	Loose frame or bad locating	Use registration guides and tight frame

Module-wise expected questions

- List screen printing equipment and materials.
- Explain photographic stencil methods.
- Compare monofilament and multifilament screens.
- Explain squeegee selection.
- State applications of screen printing.

MODULE 4 • CO4 • 14 HOURS

Process Selection and Troubleshooting

Assess problems, analyse them and give remedies.

Module outcome map

Module outcome	Description	Hours	Level
M4.01	Selection of printing process according to the job	5	Understanding
M4.02	Trouble shooting related to flexo	4	Understanding
M4.03	Trouble shooting related to gravure and screen printing	5	Understanding

Textbook chapter explanation

This chapter expands every syllabus topic into practical prepress / press / post-press learning points.

Procedure for selecting printing process according to job

Procedure for selecting printing process according to job

Workflow in a printing press and workflow diagram

Workflow in a printing press and workflow diagram

Flexo troubles

Flexo troubles: tracking, trapping, mottle, striations, weak colour, adhesion, foaming and smearing

Gravure troubles

Gravure troubles: abrasion, scumming, drying-in, bleeding, haze, pinholes, blocking, static and brittleness

Screen printing troubles and remedies

Screen printing troubles and remedies

Root-cause analysis

Root-cause analysis: machine, material, method, ink, operator and environment

Malayalam support: □ module □□□□□□□□□□□□□ definition □□□□□□□ □□□□□□□□□□□□ □□□□□. Process flow, machine part, job application, defect and remedy □□□□□□□ □□□□□ answer-□ □□□□□□□.

Illustration / diagram lab

Process Selection and Troubleshooting - labelled process diagram for exam answer practice.

Industrial application

Troubleshooting prevents waste, downtime and customer rejection. Production engineers must link defect appearance to process variable instead of blindly changing settings.

> Exam answer format: definition → principle → labelled diagram → operation sequence → advantages → limitations → applications → defects/remedies.

Troubleshooting table

Problem	Likely cause	Action / remedy
Mottle	Uneven ink/substrate interaction	Check ink, anilox/cylinder, pressure and substrate
Drying-in	Ink dries in cells/mesh	Adjust solvent, humidity and speed
Blocking	Printed sheets/roll layers stick	Improve drying and reduce residual solvent
Abrasion	Poor rub resistance	Select suitable ink and coating

Module-wise expected questions

- Prepare a process selection chart for a printing job.
- Explain workflow in a printing press.
- Give causes and remedies for mottle and striations.
- Give causes and remedies for scumming and pinholes.
- Explain troubleshooting procedure for screen printing.

FORMULA / RULE BANK

Rule bank, definition bank and procedure bank

This subject is process-heavy, so the most useful bank is a procedure/rule bank instead of pure numerical formulae.

Flexographic Printing Process

Rule: Always answer with principle, labelled diagram, sequence, materials/substrates, application and quality trouble.

Used in: Used for labels, flexible packaging, corrugated cartons, foil laminates and narrow web products. In industry, wrong anilox selection gives weak colour, over-inking, dot g...

Small example: Select this process only after checking job quantity, material, quality, cost and machine availability.

Gravure Printing Process

Rule: Always answer with principle, labelled diagram, sequence, materials/substrates, application and quality trouble.

Used in: Gravure is chosen for long-run high-quality magazines, laminates, packaging and decorative products. Cylinder and doctor blade condition decide consistency, density and s...

Small example: Select this process only after checking job quantity, material, quality, cost and machine availability.

Screen Printing Process

Rule: Always answer with principle, labelled diagram, sequence, materials/substrates, application and quality trouble.

Used in: Screen printing is used for T-shirts, decals, PCBs, glass, ceramics, panels, name plates and thick ink deposit work. Squeegee hardness, mesh count and off-contact gap dec...

Small example: Select this process only after checking job quantity, material, quality, cost and machine availability.

Process Selection and Troubleshooting

Rule: Always answer with principle, labelled diagram, sequence, materials/substrates, application and quality trouble.

Used in: Troubleshooting prevents waste, downtime and customer rejection. Production engineers must link defect appearance to process variable instead of blindly changing settings...

Small example: Select this process only after checking job quantity, material, quality, cost and machine availability.

SOLVED EXAMPLES / PROBLEM LAB

Worked answer patterns

Example 1: Flexographic Printing Process

Given: A production job or defect belongs to this module.

Logic: Identify material, image carrier/tool, machine setting, operation sequence and quality requirement.

Answer: Used for labels, flexible packaging, corrugated cartons, foil laminates and narrow web products. In industry, wrong anilox selection gives weak colour, over-inking, dot gain or drying problems.

> Do not give the process name alone. Justify using job requirement and quality control.

Example 2: Gravure Printing Process

Given: A production job or defect belongs to this module.

Logic: Identify material, image carrier/tool, machine setting, operation sequence and quality requirement.

Answer: Gravure is chosen for long-run high-quality magazines, laminates, packaging and decorative products. Cylinder and doctor blade condition decide consistency, density and streak-free printing.

> Do not give the process name alone. Justify using job requirement and quality control.

Example 3: Screen Printing Process

Given: A production job or defect belongs to this module.

Logic: Identify material, image carrier/tool, machine setting, operation sequence and quality requirement.

Answer: Screen printing is used for T-shirts, decals, PCBs, glass, ceramics, panels, name plates and thick ink deposit work. Squeegee hardness, mesh count and off-contact gap decide print quality.

> Do not give the process name alone. Justify using job requirement and quality control.

Example 4: Process Selection and Troubleshooting

Given: A production job or defect belongs to this module.

Logic: Identify material, image carrier/tool, machine setting, operation sequence and quality requirement.

Answer: Troubleshooting prevents waste, downtime and customer rejection. Production engineers must link defect appearance to process variable instead of blindly changing settings.

> Do not give the process name alone. Justify using job requirement and quality control.

PRACTICE AND QUICK REVISION

Last-day revision, expected questions and objective practice

One-line revision points

- Flexographic Printing Process: Explain the flexographic printing process.
- Gravure Printing Process: Explain and comprehend the gravure printing process.
- Screen Printing Process: Explain and comprehend the screen-printing process.
- Process Selection and Troubleshooting: Assess problems, analyse them and give remedies.

Common exam traps

- No labelled diagram.
- No operation sequence.
- Confusing machine part and process result.
- Writing advantages without limitations.
- Not connecting defects with remedies.

Expected questions

CO1: List main sections of a flexographic press.

CO1: Draw a flexo printing unit and label anilox, plate and impression cylinder.

CO1: Explain chambered doctor blade inking system.

CO1: Compare inline, stack and CI flexo presses.

CO1: State the importance of corona treatment.

CO2: Draw a gravure printing unit.

CO2: Differentiate static and dynamic balancing.

CO2: Explain gravure cylinder manufacture methods.

CO2: List doctor blade types and problems.

CO2: Explain solvent recovery in gravure drying.

CO3: List screen printing equipment and materials.

CO3: Explain photographic stencil methods.

CO3: Compare monofilament and multifilament screens.

CO3: Explain squeegee selection.

CO3: State applications of screen printing.

CO4: Prepare a process selection chart for a printing job.

CO4: Explain workflow in a printing press.

CO4: Give causes and remedies for mottle and striations.

CO4: Give causes and remedies for scumming and pinholes.

CO4: Explain troubleshooting procedure for screen printing.

Objective questions

1. Which answer format is most suitable for Flexographic Printing Process?

A. Definition only
B. Only advantages
C. Principle, diagram, operation, application and defects
D. Only history

Answer: C

2. Which answer format is most suitable for Gravure Printing Process?

A. Definition only
B. Only advantages
C. Principle, diagram, operation, application and defects
D. Only history

Answer: C

3. Which answer format is most suitable for Screen Printing Process?

A. Definition only
B. Only advantages
C. Principle, diagram, operation, application and defects
D. Only history

Answer: C

4. Which answer format is most suitable for Process Selection and Troubleshooting?

A. Definition only
B. Only advantages
C. Principle, diagram, operation, application and defects
D. Only history

Answer: C

FRESH MODEL QUESTION PAPER

4101 - Mechanism of Printing Machines II

Time: 3 Hours • Maximum Marks: 100 • Internal choice may be applied by teachers. This is a fresh practice paper, not copied from official questions.

Course Code: 4101
Course Title: Mechanism of Printing Machines II
Time: 3 Hours
Maximum Marks: 100

Part A - Short Answer Questions
1. Define Mechanism of Printing Machines II and state its industrial relevance.
2. Explain the prerequisite knowledge required for this subject.
3. List main sections of a flexographic press.
4. Draw a flexo printing unit and label anilox, plate and impression cylinder.
5. Explain chambered doctor blade inking system.
6. Compare inline, stack and CI flexo presses.
7. Draw a gravure printing unit.
8. Differentiate static and dynamic balancing.
9. Explain gravure cylinder manufacture methods.
10. List doctor blade types and problems.
11. List screen printing equipment and materials.
12. Explain photographic stencil methods.
13. Compare monofilament and multifilament screens.
14. Explain squeegee selection.
15. Prepare a process selection chart for a printing job.
16. Explain workflow in a printing press.
17. Give causes and remedies for mottle and striations.
18. Give causes and remedies for scumming and pinholes.

Part B - Descriptive / Diagram Questions
Answer any five. Each answer must include a labelled diagram wherever applicable.

Part C - Application / Troubleshooting
Prepare process selection and defect-remedy answers from practical production situations.

COMPLETE MODEL ANSWERS

Full answer guide, keywords and mistakes to avoid

CO1 answer guide - Flexographic Printing Process

Write the definition, principle, labelled diagram, working sequence, main parts/materials, advantages, limitations, applications and quality problems. Important keywords: Principle, advantages, limitations and characteristics of flexography, Infeed, printing, inking, plate carrier, impression cylinder, drying and outfeed sections, Flexo configurations: inline, stack and central impression, Rubber, photopolymer, metal backed and magnetic image carriers, Rubber plate, sheet photopolymer and liquid photopolymer preparation.

Common mistakes: no diagram, no sequence, wrong terminology, no industrial application and no remedy for defects.

CO2 answer guide - Gravure Printing Process

Write the definition, principle, labelled diagram, working sequence, main parts/materials, advantages, limitations, applications and quality problems. Important keywords: Principle, advantages, limitations and characteristics of gravure, Infeed, printing, drying and outfeed sections, Gravure cylinder parts: axis, shaft, diameter, circumference and face length, Static and dynamic balancing, Gravure well configuration and effect on tone transfer.

Common mistakes: no diagram, no sequence, wrong terminology, no industrial application and no remedy for defects.

CO3 answer guide - Screen Printing Process

Write the definition, principle, labelled diagram, working sequence, main parts/materials, advantages, limitations, applications and quality problems. Important keywords: Screen printing definition and image transfer principle, Frames: wooden, metal and self-stretching frames, Mesh materials: silk, synthetic and stainless steel, Stencil types: hand-cut, tusche and glue, photographic, Screen fabrics: multifilament and monofilament.

Common mistakes: no diagram, no sequence, wrong terminology, no industrial application and no remedy for defects.

CO4 answer guide - Process Selection and Troubleshooting

Write the definition, principle, labelled diagram, working sequence, main parts/materials, advantages, limitations, applications and quality problems. Important keywords: Procedure for selecting printing process according to job, Workflow in a printing press and workflow diagram, Flexo troubles: tracking, trapping, mottle, striations, weak colour, adhesion, foaming and smearing, Gravure troubles: abrasion, scumming, drying-in, bleeding, haze, pinholes, blocking, static and brittleness, Screen printing troubles and remedies.

Common mistakes: no diagram, no sequence, wrong terminology, no industrial application and no remedy for defects.

REFERENCES

Official syllabus references and online resources

Text / Reference

- T1 Gravure Process and Technology, GAA.
- R1 Flexographic Principles and Practices, 4th and 5th edition, Foundation of Flexographic Technical Association.
- R2 Handbook of Printing and Production, Michael Bernard, John Peacock.
- R3 Printing Technology 5th edition, Adams J.M, Delmar Publishers, New York.
- R4 Hand book of Print Media, Helmut Kipphan, Springer Science & Business Media.

Online resources

- www.gravurecylinders.co.za/the-process
- https://www.thebookdesigner.com
- https://www.colormatters.com
- https://www.britannica.com/technology/flexography
- https://medium.muz.li/a-beginners-guide-to-screen-printing-by-a-complete-beginner
- https://www.tech-sleeves.com/the-company/flexography